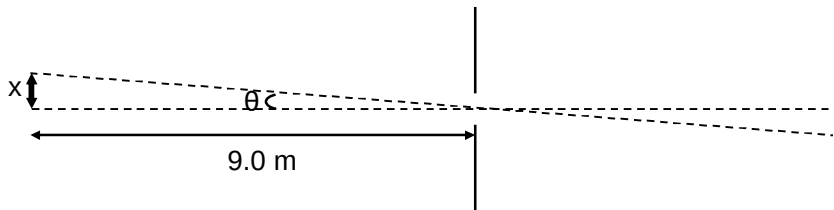


$$v = 3.0 \times 300 = 900 \text{ m s}^{-1}$$

16

Ans : B



Using Rayleigh's criterion, the minimum angle for image to be just resolved, $\sin \theta = \frac{\lambda}{b}$

$$\sin \theta \approx \tan \theta$$

$$\frac{\lambda}{b} = \frac{x}{9.0}$$

$$\tan \theta = x / 9.0 = 2.0 \times 10^{-5}$$

$$x = 1.8 \times 10^{-4} = 0.18 \text{ mm}$$

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Ans: C